

CALRIG · DESIGN DECISION RECORD

CalRig TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	CLR-DDR-001	Version	0.1
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/calrig		
Source	docs/decisions/0001-trl2-review-decisions.md @ a22ff73		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review recommendations adopted for TRL 3 work under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** proposed. The recommendation for each item A1 to A6 is adopted for TRL 3 work pending Amish's review; item O1 remains open.

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed seven items as "Proposed, awaiting Amish". On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He did not review this batch item by item. Every item that carried a recommendation is therefore adopted as recommended for TRL 3 work, open for his review. Items without a recommendation stay open. None of these entries is a decision by Amish.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in CLR-PRC-001 v0.2, Open questions.

■ Decision

Table 1. Items adopted for TRL 3 work.

#	Item	Recommendation adopted	Status
A1	Budget	Option (a): raise <code>budget_usd</code> from \$300 to \$400. <code>budget_usd</code> in <code>project.yaml</code> is not changed; \$400 is recorded here and in the review note as awaiting Amish, and CLR-CAL-001 states the cost against both figures (\$404 in parts, over both).	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. The budget figure itself remains awaiting Amish.
A2	Reference strategy	Transfer references (two SHT45, SCD30, collocated SPS30) checked against salt fixed points, an ice point and field collocation, with the uncertainty chain stated in every report.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review.
A3	Cooling limit (R1)	Accept 10 °C only in rooms at 25 °C or below for the first build; no second module or ice-water exchanger. R1 is redefined accordingly in CLR-REQ-001 v0.3.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review.
A4	Test aerosol	Incense smoke decay; every report states the aerosol type.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review.
A5	Gas scope	NO2 by field collocation only for now; no gas cylinder, dilution or toxic gas handling in CalRig. AirStreet's README already states that NO2 is calibrated by field collocation; its pitch line still says "calibrated on CalRig" and is AirStreet's to change. AirStreet is not edited from this repo.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review.
A6	Door condensation	Insulated removable door panel for hot, humid points (and for cold points).	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review.

Table 2. Items that remain open.

#	Item	Status
O1	Collocation partner for the transfer SPS30 (regulatory monitoring station, university site or an AQ-SPEC style program). No recommendation was made.	Proposed, awaiting Amish

The pitch and problem lines in `project.yaml` were not flagged for rewording in the review and are unchanged.

■ Consequences

- CLR-REQ-001 v0.3: R1 redefined to 10 to 40 °C in rooms of 15 to 25 °C, with the lowest reachable set point stated in each record for rooms up to 30 °C (A3). R12 keeps the \$300 target from `project.yaml` and notes the proposed \$400.
- CLR-PRC-001 v0.3: the key design choices above are no longer shown as "proposed"; they are adopted for TRL 3 work, open for review. The door panel becomes part of the jacket (A6).
- CLR-PRB-001 v0.3: the constraint and out-of-scope lines reflect A1 and A5.
- CLR-CAL-001 v0.1 checks every requirement against the adopted design. It found that R2 (20 °C at 85 % RH in the reference room), R8 (O1 still open), R12 (cost) and R13 (mass) are not met. The design changes it made (full jacket, 120 mm mixing fan, sleeved bubbler with a trace-heated line, 600 x 500 mm base) are engineering

changes within the adopted concept. Further options, such as a larger inner sink, are proposed in `docs/REVIEW.md` and awaiting Amish.